Kinematics of Mechanisms - Homework 1

Report requirements

Report (solution) must contain the following points:

1. The Cinderella Introduction and DOF tasks:
    1. Print screen of each mechanisms created via Cinderella Software with written value of the mobility (DOF). You can write mobility in the Cinderella above your mechanisms using tool "Add text".
    2. The files (.cdy format) from Cinderella with these mechanisms.
2. The other tasks:
    1. Full and detailed explanation of the solution as PDF-file.
    2. Fill the answer table (find the example in the end of the file)

Report and .cdy-files should be placed in one archive and uploaded to Moodle.

> ⓘ **Zip file Requirements**
>
> The zip file name should include your name and ID number in the following format: **"HW1_firstnameLASTNAME_xxxxxxxxx.zip"**. Where **firstname** is your first name, **LASTNAME** is your last name in capital letters and **xxxxxxxxx** is your ID number.

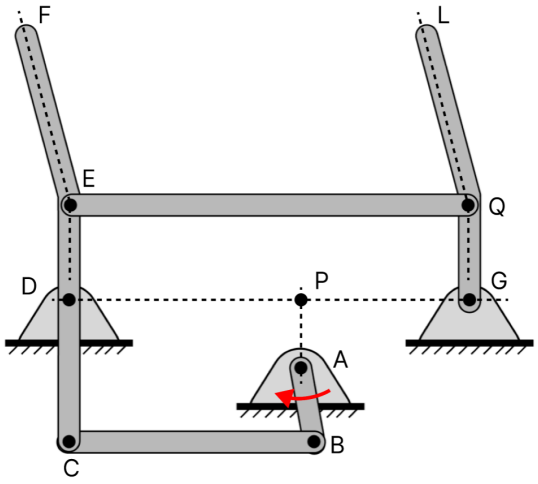
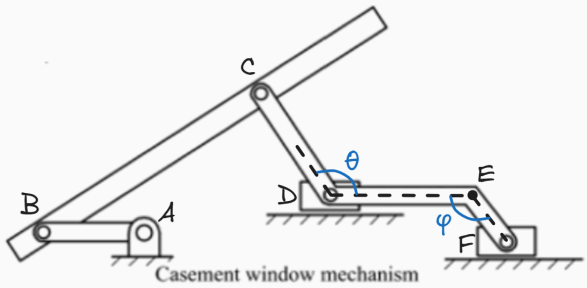
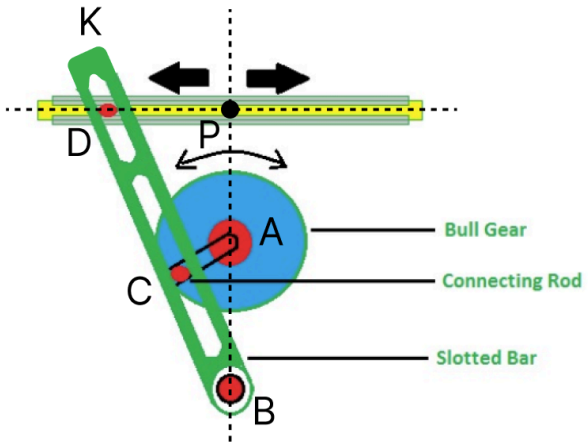
Question

1. Explain the difference between Mechanism and Machine
2. What is Four-bar linkage and which types of four-bar linkage are exist (Explain their difference)?
3. Describe all main lower kinematics pairs with their DOF values.
4. What is difference among Grashof Mechanism, Non-Groshof Mechanism and Special Case?

Tasks

Part A: The Cinderella Introduction and DOF

You must create these mechanisms in the Cinderella Software. Define the degree of freedom (DOF) for these mechanisms. You must define and write number of links, joints with DOF = 1 and joints with DOF = 2.

Mechanism Image	Given
	AB = 2, BC = 9, CD = 6, DE = 6 , EQ = 16, QG = 6, EF = 8, QL = 8, DP = 9, PG = 7, PA = 4, ∠DEF = 15°, ∠LQG = 15°
	AB = 4, BC = 8, CD = 5, DE = 7, EF = 5 The angles θ and φ are fixed. θ = 125° φ = 125° The point D and point F are sliders.
	AC = 4, AB = 8, AP = 8, KB = 20

Part B: Grashof condition and Linkage classification

Task 1

Determine the class of the mechanism according to Grashof's condition and identify which types of mechanisms can be constructed. (All of these mechanism are four-bar linkage).

Case 1	Case 2	Case 3
$l_1 = 12\text{cm}, \quad l_2 = 4\ \text{cm}$ $l_3 = 10\ \text{cm}, \quad l_4 = 9\ \text{cm}$	$l_1 = 4\ \text{cm}, \quad l_2 = 8\ \text{cm}$ $l_3 = 11\ \text{cm}, \quad l_4 = 7\ \text{cm}$	$l_1 = 6\ \text{cm}, \quad l_2 = 3\ \text{cm}$ $l_3 = 5\ \text{cm}, \quad l_4 = 4\ \text{cm}$

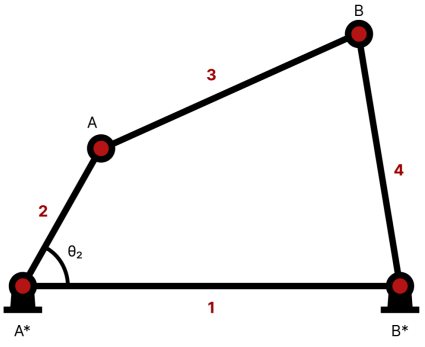
Task 2

You are designing a four-bar mechanism with a fixed ground link $l_1 = 4\ \text{cm}$ and an input link $l_2 = 9\ \text{cm}$.

1. Determine integer values for the coupler length (l_3) and output link length (l_4) so that the mechanism functions as a **Double-Crank (Drag Link)** mechanism.
2. Explain your choice using Grashof's Law condition and state which link must be the shortest (S) for this mechanism type

Part C: Position analysis of a planar Four-bar linkage

A four-bar mechanism is given with two fixed pivots, A^* and B^*. The following values are known: the frame length A^*B^*, the input link length A^*A the coupler length AB, the output link length B^*B, and the input angle θ_2 measured from the positive x-axis. You need to define main diagonal line between points A^* and B, angels between the other links and positive x-axis (θ_3, θ_4). You need to use numerical method.

Image	Task 1	Task 2
	A four-bar mechanism with fixed pivots A^* and B^*. Given: $r_1 = 90$ $r_2 = 40$ $r_3 = 140$ $r_4 = 110$ $\theta_2 = 55^\circ$ Define: $AB^* = L$, θ_3, θ_4	A four-bar mechanism with fixed pivots A^* and B^*. Given: $r_1 = 70$ $r_2 = 55$ $r_3 = 120$ $r_4 = 95$ $\theta_2 = 115^\circ$ Define: $AB^* = L$, θ_3, θ_4

Example of an answer table (for tasks)

Part A			Part B				Part C	
Case 1	Case 2	Case 3	Task 1			Task 2	Case 1	Case 2
			Case 1	Case 2	Case 3			